

AVERAGES

AVERAGE: In general average is calculated by representing a group of values with a single value and it is defined as

$$\text{Average} = \frac{\text{Sum of all observations}}{\text{No. of observations}}$$

Note: Average of any number of quantities will always be greater than the lowest value and lesser than the highest value.

Average Speed:

❖ Basically average speed is depends on 2 conditions

1) Distances are equal

2) Distances are not equal

1) Distances are equal:

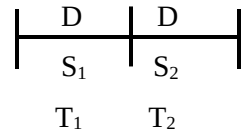
✓ If there are 2 equal distances,

$S_1 \rightarrow$ Speed during first half of the distance

$S_2 \rightarrow$ Speed during second half of the distance

$T_1 \rightarrow$ Time taken to cover first half of the distance

$T_2 \rightarrow$ Time taken to cover second half of the distance



$$\text{Average Speed} = \frac{\text{Total distance}}{\text{Total time}} = \frac{D + D}{T_1 + T_2}$$

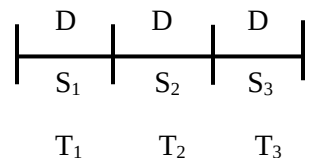
$$\text{Where } T_1 = \frac{D}{S_1}, T_2 = \frac{D}{S_2} \quad \because S = \frac{D}{T} \quad \Rightarrow \quad T = \frac{D}{S}$$

$$\text{Average Speed} = \frac{2D}{\frac{D}{S_1} + \frac{D}{S_2}} = \frac{2D}{D[\frac{1}{S_1} + \frac{1}{S_2}]} = \frac{2S_1S_2}{S_1 + S_2}$$

$$\text{Average speed} = \frac{2S_1S_2}{S_1 + S_2}$$

✓ Similarly, if there are 3 equal distances

$$\text{Average Speed} = \frac{\text{Total distance}}{\text{Total time}}$$



$$\text{Average Speed} = \frac{D + D + D}{T_1 + T_2 + T_3} = \frac{3D}{\frac{D}{S_1} + \frac{D}{S_2} + \frac{D}{S_3}} = \frac{3D}{D[\frac{1}{S_1} + \frac{1}{S_2} + \frac{1}{S_3}]}$$

$$\text{Average speed} = \frac{3S_1S_2S_3}{S_1S_2 + S_2S_3 + S_3S_1}$$

2) Distances are not equal:

- ✓ If distances are not equal, then

$$\text{Average Speed} = \frac{\text{Total distance}}{\text{Total time}}$$

D_1	D_2	D_3
S_1	S_2	S_3
T_1	T_2	T_3

$$\text{Average speed} = \frac{D_1 + D_2 + D_3}{T_1 + T_2 + T_3}$$

Some important formulae to be remember:

- ✓ Sum of 'n' natural numbers = $\frac{n(n+1)}{2}$
- ✓ Sum of squares of 'n' natural numbers = $\frac{n(n+1)(2n+1)}{6}$
- ✓ Sum of cubes of 'n' natural numbers = $\frac{n^2(n+1)^2}{4}$ or $\left[\frac{n(n+1)}{2}\right]^2$
- ✓ a) The average of first 'n' consecutive odd numbers is 'n'.
- b) The average of first 'n' consecutive even numbers is 'n+1'.
- ✓ a) Average of odd numbers from 1 to n is

$$\text{Average} = \frac{\text{last odd number} + 1}{2}$$

- b) Average of even numbers from 1 to n is

$$\text{Average} = \frac{\text{last even number} + 2}{2}$$

- ✓ If the difference between any two consecutive numbers is same, then

$$\text{Average} = \frac{\text{First number} + \text{last number}}{2}$$

- ✓ The average of 'n' numbers is A.
- a) if each number is multiplied by 'k', then new average = $A \times k$
- b) if each number is added by 'k', then new average = $A + k$
- c) if each number is subtracted by 'k', then new average = $A - k$
- d) if each number is divided by 'k', then new average = $A \div k$

SOLVED EXAMPLES

Q - 1

What is the average of all numbers 47, 59, 69, 57, 79, 73?

Sol:
$$\text{Average} = \frac{\text{Sum of observations}}{\text{No. of observations}} = \frac{47 + 59 + 69 + 57 + 79 + 73}{6} = 64$$

Alternate method:

- ✓ By observing all the given numbers, take one approximate average round figure number which is in between lowest and highest numbers.
- ✓ Consider 60 is the approximate average round figure number.
- ✓ Now, compare every number with approximate average. If the number is less take -ve sign, if the number is more take +ve sign.

$$\therefore \text{Average} = 60 + \frac{(-13 - 1 + 9 - 3 + 19 + 13)}{6} = 60 + \frac{24}{6} = 64.$$

Q - 2

Find the average of all prime numbers between 40 and 70?

Sol: Prime numbers between 40 and 70 = 41, 43, 47, 53, 59, 61, 67

$$\text{Average} = \frac{\text{Sum of observations}}{\text{No. of observations}} = \frac{41 + 43 + 47 + 53 + 59 + 61 + 67}{7} = 53.$$

Alternate method:

Consider approximate average = 50, then compare every number with 50

$$\therefore \text{Average} = 50 + \frac{(-9 - 7 - 3 + 3 + 9 + 11 + 17)}{7} = 50 + \frac{21}{7} = 53.$$

Q - 3

Find the average of 73 natural numbers?

Sol:
$$\text{Average} = \frac{\text{Sum of 73 natural numbers}}{\text{No. of observations}} \quad \therefore \text{Sum of 'n' natural no's} = \frac{n(n+1)}{2}$$

$$\text{Sum of 73 natural numbers} = \frac{73(73+1)}{2} = \frac{73 \times 74}{2}$$

$$\text{Average} = \frac{\frac{73 \times 74}{2}}{73} = 37.$$

Shortcut:

Average of 1, 2, 3,73

For the above numbers difference between any two consecutive numbers is same, then

$$\text{Average} = \frac{\text{First number} + \text{Last number}}{2}$$

$$\therefore \text{Average} = \frac{1 + 73}{2} = 37.$$

Q - 4**Find the average of squares of natural numbers from 1 to 35.**

Sol: Sum of squares of 'n' natural numbers = $\frac{n(n+1)(2n+1)}{6}$

Sum of squares from 1 – 35 = $\frac{35 \times 36 \times 71}{6}$ $\therefore n = 35$

$\therefore$ Average = $\frac{\frac{35 \times 36 \times 71}{6}}{35} = \frac{36 \times 71}{6} = 426.$

Q - 5**What is the average of cubes of natural numbers from 1 to 31?**

Sol: Sum of cubes of 'n' natural numbers = $\frac{n^2(n+1)^2}{4}$

Sum of cubes from 1 – 31 = $\frac{31^2 \times 32^2}{4}$

$\therefore$ Average = $\frac{\text{Sum of observations}}{\text{No. of observations}} = \frac{\frac{31^2 \times 32^2}{4}}{31} = \frac{31 \times 32^2}{4} = 7936.$

Q - 6**Find the average of first 80 consecutive odd numbers.**

Sol: Average of first 'n' consecutive odd numbers = n

Here n = 80, therefore Average = 80.

Q - 7**Find the average of first 53 consecutive even numbers.**

Sol: Average of first 'n' consecutive even numbers = n + 1

Here n = 53, therefore Average = 53 + 1 = 54.

Q - 8**What is the average of odd numbers from****a) 1 to 65 b) 1 to 74**

Sol: Average of odd numbers from 1 to n = $\frac{\text{last odd number} + 1}{2}$

a) Average of odd numbers from 1 to 65 = $\frac{65 + 1}{2} = 33.$

b) Average of odd numbers from 1 to 74 = $\frac{73 + 1}{2} = 37.$

 $\therefore$ Last odd number from 1 to 74 is 73**Q - 9****What is the average of even numbers from****a) 1 to 84 b) 1 to 57**

Sol: Average of even numbers from 1 to n = $\frac{\text{Last even number} + 2}{2}$

a) Average of even numbers from 1 to 84 = $\frac{84 + 2}{2} = 43.$

b) Average of even numbers from 1 to 57 = $\frac{56 + 2}{2} = 29.$

 $\therefore$ Last even number from 1 to 57 is 56

Q - 10

Find the average of first 16 multiples of 8.

Sol:
$$\text{Average} = \frac{\text{Sum of observations}}{\text{No. of observations}} = \frac{8[1 + 2 + \dots + 16]}{16}$$

$$\text{Sum of 16 natural numbers} = \frac{16 \times 17}{2} \quad \therefore \text{Sum of 'n' natural numbers} = \frac{n(n+1)}{2}$$

$$\therefore \text{Average} = \frac{8\left[\frac{16 \times 17}{2}\right]}{16} = 68.$$

Shortcut:

First 16 multiples of 8 = 8, 16, 24, ..., 120, 128

Since, the difference between any two consecutive numbers is same, then

$$\text{Average} = \frac{\text{First number} + \text{Last number}}{2}$$

$$\therefore \text{Average} = \frac{8 + 128}{2} = 68.$$

Q - 11

Find the average of multiples of 16 in the natural number series from 150 to 350.

Sol: First multiple of 16 after 150 = 160, Last multiple of 16 before 350 = 336

Average of numbers 160, 176, ..., 336

$\therefore$ The difference between any two consecutive numbers is same, then

$$\therefore \text{Average} = \frac{\text{First number} + \text{Last number}}{2} = \frac{160 + 336}{2} = 248.$$

Q - 12

The average of 15 numbers is 9. What will be the new average, if each number is multiplied by 12?

Sol:

Note: The average of 'n' numbers is A. If each number is multiplied by 'k', then

$$\text{New average} = A \times k$$

Here, initial average A = 9 and k = 12

$$\therefore \text{New average} = 9 \times 12 = 108.$$

Q - 13

There are 40 boys and 50 girls in a class. If the average weight of boys is 54 kg and the average weight of girls is 45 kg. What is the average weight of entire class?

Sol:
$$\text{Average weight of entire class} = \frac{\text{Sum of boys} + \text{Sum of girls}}{\text{Total strength}}$$

$$\text{Sum of boys} = \text{Average of boys} \times \text{No. of boys} = 54 \times 40 = 2160 \text{ kg}$$

Sum of girls = Average of girls $\times$ No. of girls = $45 \times 50 = 2250$ kg

$$\therefore \text{Average weight of entire class} = \frac{2160 + 2250}{40 + 50} = 49 \text{ kg.}$$

Q - 14

Naresh bought 15 pens, 6 books and 9 pencil boxes each at a rate of Rs. 12, Rs. 25 and Rs. 10 respectively. Find the average cost of all the items.

Sol:

$$\text{Average cost of all items} = \frac{\text{Total cost of all items together}}{\text{No. of items}}$$

$$\therefore \text{Average} = \frac{15 \times 12 + 6 \times 25 + 9 \times 10}{15 + 6 + 9} = \text{Rs. } 14.$$

Q - 15

If the average of 8 consecutive odd numbers is 32. Find the largest of these numbers.

Sol:

Consider, 8 consecutive odd numbers are $x, x + 2, x + 4, x + 6, x + 8, x + 10, x + 12, x + 14$

$\therefore$ Gap between any two consecutive odd numbers is 2

$$\text{Average} = \frac{\text{Sum of observations}}{\text{No. of observations}} = \frac{x + x + 2 + x + 4 + x + 6 + x + 8 + x + 10 + x + 12 + x + 14}{8}$$

$$32 = \frac{8x + 56}{8} \Rightarrow 8x = 32 \times 8 - 56 = 200 \Rightarrow x = 25$$

$$\therefore \text{Largest number} = x + 14 = 25 + 14 = 39.$$

Alternate method:

Consider, 8 consecutive odd numbers are $x - 7, x - 5, x - 3, x - 1, x + 1, x + 3, x + 5, x + 7$

$$\text{Average} = \frac{x - 7 + x - 5 + x - 3 + x - 1 + x + 1 + x + 3 + x + 5 + x + 7}{8}$$

$$32 = \frac{8x}{8} \Rightarrow x = 32$$

$$\therefore \text{Largest number} = x + 7 = 32 + 7 = 39.$$

Shortcut:

✓ Average means middle number. Since 32 is an even number therefore 32 is not one of the 8 consecutive odd numbers.

25	27	29	31	33	35	37	39
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Left side 4 consecutive odd no's $\leftarrow$ 32
Avg. $\rightarrow$ Right side 4 consecutive odd no's

$$\therefore \text{Largest number} = 39.$$

Q - 16

A boy is going to school from his house at a speed of 30 kmph and return home at a speed of 45 kmph. Find the average speed.

Sol:

In the given problem, distances are equal in both the cases.

$$\text{Average speed} = \frac{2S_1S_2}{S_1 + S_2}$$

$$\therefore \text{Average speed} = \frac{2 \times 30 \times 45}{30 + 45} = 36 \text{ kmph.}$$

Q - 17

A man covers three equal distances at a speeds of 15 kmph, 20 kmph and 30 kmph respectively, then find the average speed during the entire journey.

Sol: According to question, man covers 3 equal distances.

$$\text{Then, Average speed} = \frac{3S_1S_2S_3}{S_1S_2 + S_2S_3 + S_3S_1}$$

$$\therefore \text{Average speed} = \frac{3 \times 15 \times 20 \times 30}{15 \times 20 + 20 \times 30 + 30 \times 15} = 20 \text{ kmph.}$$

Q - 18

An aeroplane travels distances 1200 km, 1500 km and 2000 km with a speeds of 600 kmph, 500 kmph and 400 kmph respectively. What is the average speed of entire journey?

Sol: In the given problem, distances are not equal.

$$\text{Then, Average Speed} = \frac{\text{Total distance}}{\text{Total time}} = \frac{D_1 + D_2 + D_3}{T_1 + T_2 + T_3}$$

$$T_1 = \frac{D_1}{S_1} = \frac{1200}{600} = 2 \text{ hours, } T_2 = \frac{D_2}{S_2} = \frac{1500}{500} = 3 \text{ hours, } T_3 = \frac{D_3}{S_3} = \frac{2000}{400} = 5 \text{ hours}$$

$$\therefore \text{Average Speed} = \frac{1200 + 1500 + 2000}{2 + 3 + 5} = 470 \text{ kmph.}$$

Q - 19

The average of 11 consecutive even numbers is 80, then what is the difference between smallest and largest numbers?

Sol: Consider, 11 consecutive even numbers are

$$x - 10, x - 8, x - 6, x - 4, x - 2, x, x + 2, x + 4, x + 6, x + 8, x + 10$$

$$\text{Average} = \frac{\text{Sum of observations}}{\text{No. of observations}} \Rightarrow 80 = \frac{11x}{11} \Rightarrow x = 80$$

$$\text{Smallest number} = x - 10 = 80 - 10 = 70 \quad \& \quad \text{Largest number} = x + 10 = 80 + 10 = 90$$

$$\therefore \text{Difference between smallest and largest numbers} = 90 - 70 = 20.$$

Shortcut – 1:

✓ Average means middle number. Since 80 is an even number therefore 80 is one of the 11 consecutive even numbers which is middle number.

70	72	74	76	78	80	82	84	86	88	90
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Left side 5 consecutive even no's ← Avg. → Right side 5 consecutive even no's

Smallest number = 70, Largest number = 90

$$\therefore \text{Difference} = 90 - 70 = 20.$$

Shortcut – 2:**Note:**

- ✓ If there are n consecutive numbers, then $(n - 1)$ gaps are formed between smallest and largest numbers.
- ✓ The gap between any two consecutive even numbers (or) odd numbers is always '2'.

❖ There are 11 consecutive even numbers, therefore 10 gaps are formed between smallest and largest numbers.

$$\therefore \text{Difference} = 10 \text{ gaps} = 10 \times 2 = 20.$$

$$\therefore \text{Each gap} = 2$$

Q - 20

The average height of A and B is 155 cm, that of B and C is 162 cm and the average height of C and A is 170 cm. Find the height of A.

Sol:

Sum of observations = Average $\times$ No. of observations

$$A + B = 155 \times 2 = 310 \text{ cm} \quad \rightarrow \quad (1)$$

$$B + C = 162 \times 2 = 324 \text{ cm} \quad \rightarrow \quad (2)$$

$$C + A = 170 \times 2 = 340 \text{ cm} \quad \rightarrow \quad (3)$$

$$(1) + (2) + (3) \Rightarrow 2(A + B + C) = 310 + 324 + 340 = 974 \text{ cm}$$

$$A + B + C = 487 \text{ cm}$$

$$\therefore \text{Height of A} = (A + B + C) - (B + C) = 487 - 324 = 163 \text{ cm.}$$

Q - 21

The average of 5 consecutive odd numbers A, B, C, D and E is 53. What is the product of B and E?

Sol:

Consider, 5 consecutive odd numbers are

$x - 4$	$x - 2$	x	$x + 2$	$x + 4$
A	B	C	D	E

$$\text{Average} = \frac{x - 4 + x - 2 + x + x + 2 + x + 4}{5} = 53 \quad \Rightarrow \quad x = 53$$

$$B = x - 2 = 53 - 2 = 51 \text{ \& } E = x + 4 = 53 + 4 = 57$$

$$\therefore \text{Product of B and E} = 51 \times 57 = 2907.$$

Shortcut:

- ✓ Average means middle number. Since 53 is an odd number, therefore 53 is one of the 5 consecutive odd numbers which is middle number.

A	B	C	D	E
49	51	53	55	57

∴ Product of B and E = $51 \times 57 = 2907$.

Q - 22

The average salary of entire staff in an office is Rs. 600 per day. The average salary of officers is Rs. 1500 and that of non – officers is Rs. 400. Find the number of non – officers, if the number of officers are 22.

Sol: Consider, No. of non – officers = x

$$\text{Average of entire staff} = \frac{\text{Sum of officers} + \text{Sum of non-officers}}{\text{Total staff}}$$

$$600 = \frac{1500 \times 22 + 400 \times x}{22 + x} \Rightarrow 600 = \frac{100(15 \times 22 + 4x)}{22 + x}$$

$$132 + 6x = 330 + 4x \Rightarrow 2x = 198 \Rightarrow x = 99$$

∴ No. of non – officers = 99.

Q - 23

There are 96 students in a class. The average marks of failed students is 22 and that of passed students is 70. If the average marks in the exam for entire class is 52, then find the number of passed students.

Sol:

Total students = Passed students + Failed students
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Consider, No. of passed students = x , then

$$\text{Failed students} = 96 - x$$

$$\therefore \text{Total no. of students} = 96$$

$$\text{Average of entire class} = \frac{\text{Sum of passed students} + \text{Sum of failed students}}{\text{Total strength}}$$

$$52 = \frac{70 \times x + 22(96 - x)}{96} \Rightarrow 4992 = 70x + 2112 - 22x$$

$$\Rightarrow 48x = 2880 \Rightarrow x = 60$$

∴ No. of passed students = 60.

Q - 24

Out of four members, first is thrice the second, fourth is twice the second and third is half of first. If the average of all four numbers is 90, then find the third number.

Sol: Consider second number is ' x '

1 st	2 nd	3 rd	4 th
$3x$	x	$\frac{3x}{2}$	$2x$

$$\therefore 1^{\text{st}} - \text{thrice of } 2^{\text{nd}}, 4^{\text{th}} - \text{twice of } 2^{\text{nd}}, 3^{\text{rd}} - \text{half of } 1^{\text{st}}$$

$$\text{Average} = \frac{3x + x + \frac{3x}{2} + 2x}{4} = 90 \Rightarrow \frac{15x}{2} = 90 \times 4 \Rightarrow x = 48$$

$$\therefore \text{Third number} = \frac{3x}{2} = \frac{3 \times 48}{2} = 72.$$

Q - 25

The average of 37 observations is 21. The average of first 18 of them is 19 and that of last 18 is 20. Find the 19th observation.

Sol: Sum of 37 observations = (Sum of first 18 obs.) + 19th obs. + (Sum of last 18 obs.)
 $19^{\text{th}} \text{ obs.} = \text{Sum of 37 obs.} - (\text{Sum of first 18 obs.}) - (\text{Sum of last 18 obs.})$
 $\therefore 19^{\text{th}} \text{ observation} = 37 \times 21 - (18 \times 19) - (18 \times 20) = 75.$

Shortcut:

First 18 obs.					Last 18 obs.				
1 2 18				19th	20 21 37				
21 21 21				21	21 21 21				→ Overall avg.
-2 ↓ -2 ↓ -2 ↓					-1 ↓ -1 ↓ -1 ↓				
19 19 19					20 20 20				
_____					_____				
-2 × 18 = -36					-1 × 18 = -18				→ Distributed value
Middle observation = Overall average + Distributed value									

Note: If the middle observation is not the part of first and last half of the no's, then 'sign' of distributed value will be opposite.

For first 18 observation → Distributed value = +36

For last 18 observation → Distributed value = +18

$\therefore 19^{\text{th}} \text{ observation} = 21 + (+36 + 18) = 75.$

Q - 26

The average weight of 31 students in a class is 55 kg. If the average weight of first 16 students is 59 kg and that of last 16 student is 52 kg. What is the weight of middle student?

Sol:

																← Last 16 →				
1	2	...	15	16	17	...	30	31												
← First 16 →																				

Here 16th value is repeated twice.

Weight of middle student = (Sum of first 16) + (Sum of last 16) – (Sum of 31 students)

$\therefore \text{Weight of middle student} = 16 \times 59 + 16 \times 52 - 31 \times 55 = 71 \text{ kg.}$

Shortcut:

First 16 students				Last 16 students			
1	2		16	16	17		31
55	55		55	55	55		55
+4↓	+4↓		+4↓	-3↓	-3↓		-3↓
59	59		59	52	52		52
$+ 4 \times 16 = + 64$				$- 3 \times 16 = - 48$			

Overall average

Distributed value

Weight of middle student = Overall average + Distributed value

Note: If the middle observation is one the part of first and last half of the numbers, then 'sign' of distributed value will not change.

∴ Weight of middle student = $55 + (+64 - 48) = 71$ kg.

Q - 27

The average of 14 results is 42. It was found later that one result 94 was wrongly entered as 38. Find the new average.

Sol:

Sum of 14 results = $14 \times 42 = 588$

∴ Sum = Average $\times$ No. of obs.

New Average = $\frac{\text{New Sum}}{\text{No. of obs.}} = \frac{588 + 94 - 38}{14} = 46$.

∴ Correct no. = 94 & Wrong no. = 38

(or)

∴ New Average = $42 + \frac{94 - 38}{14} = 46$.

Use, +ve sign for correct result
-ve sign for wrong result

Q - 28

The average marks of 25 students in a class is increased by 1.8 when one of the students who got 40 marks is replaced by another student. How many marks secured by new student?

Sol:

Consider, average of 25 students = x

Sum of 25 students = $25x$

∴ Sum = Avg. $\times$ No. of observations

New Average = $\frac{\text{New Sum}}{\text{No. of Obs.}}$

$\frac{25x - 40 + S_n}{25} = x + 1.8$

∴ Average is increased by 1.8

S_n – Marks of new student

$25x - 40 + S_n = 25x + 25 \times 1.8$

$\Rightarrow S_n = 40 + 45 = 85$

∴ New Student secured 85 marks.

Shortcut:

- ✓ Average of a class increased means new student marks definitely more than replaced student i.e., 40.
- ✓ Because of replacement of one student, all 25 students marks are increased by 1.8 each.
New student marks = $40 + (25 \times 1.8) = 85$.

Q - 29

The average wages of 15 workers are increased by Rs. 300 when two of them whose wages are Rs. 2600 and Rs. 3400 replaced by two new workers. Find the average wages of two new workers.

Sol:

Consider, average of 15 workers = x , then

Sum of 15 workers = $15x$

∴ Sum = Avg. $\times$ No. of observations

$$\text{New Average} = \frac{\text{New Sum}}{\text{No. of Obs}} \Rightarrow \frac{15x - 2600 - 3400 + W_1 + W_2}{15}$$

Where $W_1, W_2 \rightarrow$ New workers wages

∴ Average increased by 300

$$15x - 6000 + W_1 + W_2 = 15x + 4500 \Rightarrow W_1 + W_2 = 10500$$

$$\therefore \text{Average} = \frac{W_1 + W_2}{2} = \frac{10500}{2} = \text{Rs. } 5250.$$

Shortcut:

- ✓ Average of workers increased means sum of new workers wages definitely more than the sum of replaced workers i.e., $2600 + 3400 = 6000$.
- ✓ Because of replacement of 2 workers, all 15 workers wages are increased by Rs. 300 each.

$$\text{Sum of new workers} = 6000 + (15 \times 300) = 10500$$

$$\therefore \text{Average} = \frac{\text{Sum of new workers}}{2} = \frac{10500}{2} = \text{Rs. } 5250.$$

Q - 30

The average weight of 8 persons sitting in a boat is 43 kg. If the weight of boat is also included, then the average is increased to 55 kg. What is the weight of the boat?

Sol:

Sum of 8 persons = $43 \times 8 = 344$ kg

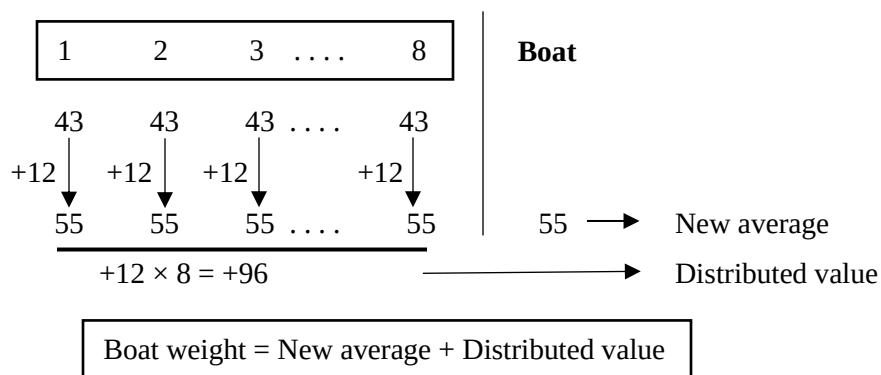
∴ Sum = Avg. $\times$ No. of observations

Sum of 8 persons and boat = $55 \times 9 = 495$ kg

Weight of boat = Sum of 8 persons and boat – Sum of 8 persons

$$\therefore \text{Weight of boat} = 495 - 344 = 151 \text{ kg.}$$

Shortcut:



Shortcut:

	1 2 20	21st innings
Initial average =	x x x	128 runs
	$+4 \downarrow$ $+4 \downarrow$ $+4 \downarrow$	
New average =	$x + 4$ $x + 4$ $x + 4$	$x + 4$
	$+4 \times 20 = +80$	Distributed value
	Runs in 21st innings = New average + Distributed value	

$$128 = (x + 4) + 80 \quad \Rightarrow \quad x + 4 = 48$$

$$\therefore \text{Average after 21st innings} = x + 4 = 48.$$

Q - 33

The average runs of a batsman in 10 innings is 52. If the highest and lowest runs were excluded, then the average would decrease by 2. If the difference between these two innings is 74, then find the highest score.

Sol:

$$\text{Sum of 10 innings} = 52 \times 10 = 520$$

$$\therefore \text{Sum} = \text{Average} \times \text{No. of observations}$$

$$\text{Sum of 8 innings} = 520 - (H + L)$$

$$\therefore H - \text{Highest runs, } L - \text{Lowest runs}$$

$$\text{Average of 8 innings} = \frac{520 - (H + L)}{8} = 52 - 2 = 50 \quad \text{Average is decreased by 2}$$

$$520 - (H + L) = 8 \times 50 = 400 \quad \Rightarrow \quad H + L = 120 \quad \rightarrow \quad (1)$$

$$\text{Also, given that } H - L = 74 \quad \rightarrow \quad (2)$$

$$(1) + (2) \quad \Rightarrow \quad H + L = 120$$

$$H - L = 74$$

$$\hline 2H = 194$$

$$\Rightarrow \quad \text{Highest runs } H = 97.$$

Shortcut:

$$\text{Distributed value} = -2 \times 8 = -16$$

1 2 8
52 52 52
$-2 \downarrow$ $-2 \downarrow$ $-2 \downarrow$
50 50 50

9 10
52 52
Excluded

Here distributed value sign will change because of exclusion of H and L.

$$H + L = (52 + 52) + 16 = 120 \quad \rightarrow \quad (1)$$

$$\text{Also, given that } H - L = 74 \quad \rightarrow \quad (2)$$

$$(1) + (2) \quad \Rightarrow \quad H + L = 120$$

$$H - L = 74$$

$$\hline 2H = 194$$

$$\Rightarrow \quad \text{Highest runs } H = 97.$$

Q - 34

The average runs scored by a cricket team in 30 matches is 290. What average should the team maintain in the next 10 matches, so that the new average becomes 300 per match?

Sol: Sum of 30 matches = $30 \times 290 = 8700$ $\therefore$ Sum = Average $\times$ No. of obs.

Consider, average for next 10 matches = x

Sum of next 10 matches = $10x$

New average after 40 matches = $\frac{\text{Sum of 30 matches} + \text{Sum of next 10 matches}}{\text{No. of matches}}$

$$300 = \frac{870 + 10x}{40} \Rightarrow 8700 + 10x = 12000 \Rightarrow 10x = 3300$$

$\therefore$ Average for next 10 matches $x = 330$.

Shortcut:

First 30 matches				Next 10 matches			
1	2		30	1	2		10
290	290		290				
↓ +10	↓ +10		↓ +10	↓	↓		↓
300	300		300	300	300		300
Distributed value $\rightarrow +10 \times 30 = +300$				$300 \times 10 = 3000 \rightarrow$ New sum			

Sum of next 10 matches = New sum + Distributed value

$$\therefore \text{Average of next 10 matches} = \frac{\text{Sum of next 10 matches}}{\text{No. of matches}} = \frac{3000 + 300}{10} = 330.$$

Q - 35

A batsman had a certain average of runs for his 13 innings. In the 14th innings he made a score of 80 runs and thus his average of runs was decreased by 4. Find his average after 14th innings.

Sol: Consider, average after 13 innings = x , after 14 innings = $x - 4$

Sum of 13 innings = $13 \times x = 13x$ $\therefore$ Sum = Average $\times$ No. of obs.

Average after 14 innings = $\frac{\text{Sum of 13 innings} + \text{14th innings}}{\text{No. of innings}}$

$$x - 4 = \frac{13x + 80}{14} \Rightarrow 13x + 80 = 14x - 56 \Rightarrow x = 136$$

$\therefore$ Average after 14 innings = $x - 4 = 136 - 4 = 132$.

Shortcut:

13 innings				14 th innings
1	2		13	80 runs
x	x		x	
↓ -4	↓ -4		↓ -4	
$x - 4$	$x - 4$		$x - 4$	
$-4 \times 13 = -52$				$x - 4 \rightarrow$ New average
				Distributed value

Runs in 14th innings = New average + Distributed value

$$80 = -52 + (x - 4) \Rightarrow x - 4 = 132$$

$\therefore$ Average after 14th innings = $x - 4 = 132$.

Q - 36

The average score of Ramesh, Mahesh and Suresh is 63. Ramesh's score is 15 less than Ajay and 10 more than Mahesh. If Ajay scored 30 marks more than the average score of Ramesh, Mahesh and Suresh. What is the sum of Mahesh's and Suresh's scores?

Sol: Sum of Ramesh, Mahesh, Suresh $R + M + S = 63 \times 3 = 189$

Given that, $R = A - 15$, $R = M + 10$ $\therefore R$ – Ramesh, M – Mahesh

$A = 63 + 30 = 93$, $R = 93 - 15 = 78$ S – Suresh, A – Ajay

$$78 = M + 10 \Rightarrow M = 68$$

$$S = (R + M + S) - (R + M) = 189 - (78 + 68) = 43$$

$\therefore$ Sum of Mahesh's and Suresh's scores $M + S = 68 + 43 = 111$.

Q - 37

The average temperatures of the town from Monday to Thursday of a week is 53°C and the average from Tuesday to Friday of a week is 56°C. If the temperatures of the Monday and Friday are in the ratio 4 : 5, then what is the temperature on Friday?

Sol: Sum of temperatures from Monday to Thursday

$$M + T + W + Th = 53 \times 4 = 212^{\circ}\text{C} \rightarrow (1)$$

Sum of temperatures from Tuesday to Friday

$$T + W + Th + F = 56 \times 4 = 224^{\circ}\text{C} \rightarrow (2)$$

$$(2) - (1) \Rightarrow F - M = 224 - 212 = 12^{\circ}\text{C}$$

Given that, Monday : Friday = 4 : 5

Difference = 1 part $\rightarrow 12^{\circ}\text{C}$

$\therefore$ Friday = 5 parts = $5 \times 12 = 60^{\circ}\text{C}$.

Q - 38

The average price of 15 books is Rs. 24, while the average price of 12 of these books is Rs. 18. Of the remaining 3 books, if the price of first book is 40% more than second and price of third book is 20% more than second. What is the price of each of these 3 books?

Sol: Sum of 15 books = $24 \times 15 = \text{Rs. } 360$ $\therefore$ Sum = Average $\times$ No. of observations

Sum of 12 books = $18 \times 12 = \text{Rs. } 216$

Sum of remaining 3 books = $360 - 216 = \text{Rs. } 144$

Consider, second book is 100%

First : Second : Third = 140% : 100% : 120% $\therefore$ 1st – 40% more than 2nd

F : S : T = 7 : 5 : 6 3^{rd} – 20% more than 2nd

$$\text{Total sum} = 18 \text{ parts} \rightarrow 144 \Rightarrow 1 \text{ part} \rightarrow \frac{144}{18} = 8$$

$\therefore$ First book = 7 parts = $7 \times 8 = \text{Rs. } 56$, Second book = 5 parts = $5 \times 8 = \text{Rs. } 40$

Third book = 6 parts = $6 \times 8 = \text{Rs. } 48$.

Q - 39

25 members contribute a certain amount to orphanage on the occasion of Independence day. 23 of them contributes Rs. 250 each and the remaining two members contributes Rs. 23 more than the average contribution of all 25. What is the total money contributed by them?

Sol: Consider, average of all 25 members = x

Sum of 23 members = $250 \times 23 = \text{Rs. } 5750$ $\therefore$ Sum = Avg. $\times$ No. of obs.

Average of 25 members = $\frac{\text{Sum of 23 members} + \text{sum of remaining 2 members}}{\text{No. of persons}}$

$$x = \frac{5750 + (x + 23) + (x + 23)}{25} \quad \therefore \text{Remaining 2 members contribute 23 more than avg.}$$

$$25x = 5796 + 2x \Rightarrow 23x = 5796 \Rightarrow x = \text{Rs. } 252$$

$\therefore$ Total money of 25 members = $25 \times x = 25 \times 252 = \text{Rs. } 6300$.

Q - 40

The average cost of 13 apples is Rs. 20 per apple. 8 more apples with an average cost of Rs. 24 per apple are purchased and 4 apples with an average cost of Rs. 11 per apple are returned. What is the average cost per apple of all the apples?

Sol: Sum of 13 apples = $20 \times 13 = \text{Rs. } 260$ $\therefore$ Sum = Avg. $\times$ No. of obs.

Sum of additional 8 apples = $24 \times 8 = \text{Rs. } 192$

Sum of returned 4 apples = $11 \times 4 = \text{Rs. } 44$

Average = $\frac{\text{Sum of 13 apples} + \text{sum of 8 apples} - \text{Sum of 4 apples}}{\text{Total no. of apples}}$

$$\therefore \text{Average} = \frac{260 + 192 - 44}{13 + 8 - 4} = \frac{408}{17} = \text{Rs. } 24.$$

Q - 41

The average of 'n' results is 45. If one more result of value 151 is added, then the average is increased by 2. Find the value of n.

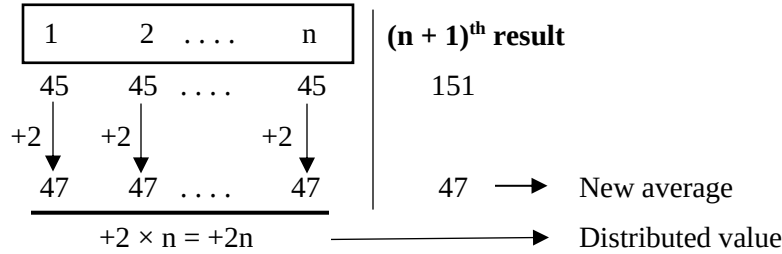
Sol: Sum of 'n' results = $45 \times n = 45n$ $\therefore$ Sum = Average $\times$ No. of observations

Average of (n + 1) results = $\frac{\text{Sum of 'n' results} + (n+1)\text{th result}}{\text{No. of results}}$

$$(45 + 2) = \frac{45n + 151}{n + 1} \quad \therefore \text{Average is increased by 2}$$

$$47n + 47 = 45n + 151 \Rightarrow 2n = 104 \Rightarrow n = 52$$

Shortcut:



$$(n + 1)^{\text{th}} \text{ result} = \text{New average} + \text{Distributed value}$$

$$151 = 47 + 2n \Rightarrow 2n = 104 \Rightarrow n = 52.$$

Q - 42

There are 56 students who go to a mess. If 8 new students are admitted, the average expenditure of mess is increased by Rs. 72 per day and the average expenditure per head is decreased by 2 rupees. What is the original expenditure of the mess?

Sol:

Consider, original expenditure per head = x

$$\text{Sum of 56 students expenses} = 56 \times x = 56x \quad \because \text{Sum} = \text{Avg.} \times \text{No. of obs.}$$

$$\text{New average} = \frac{\text{Sum of 56 students expenses} + \text{Additional expenses}}{\text{Total strength}}$$

$$x - 2 = \frac{56x + 72}{56 + 8}$$

$\because$ Average is decreased by 2

$$64x - 128 = 56x + 72 \Rightarrow x = 25$$

$$\therefore \text{Original expenditure of mess} = 56x = 56 \times 25 = \text{Rs. 1400.}$$

Q - 43

The average number of persons who are visiting central library on MON, TUE and WED is 400 and the average number of visitors on THUR, FRI is 300. If the average number of visitors per day in the week is 450, then find the average number of persons who are visiting central library in weekdays (SAT and SUN).

Sol:

$$\text{Total no. of visitors on MON, TUE, WED is } M + T + W = 400 \times 3 = 1200$$

$$\text{Total no. of visitors on THUR, FRI is } Th + F = 300 \times 2 = 600$$

$$\text{Total no. of visitors on entire week} = 450 \times 7 = 3150$$

$$\text{Average no. of visitors on weekends} = \frac{\text{Total no. of visitors on weekends}}{\text{No. of days}}$$

$$\therefore \text{Average on SAT \& SUN} = \frac{(\text{Total week}) - (M+T+W) - (Th + F)}{2} = \frac{3150 - 1200 - 600}{2} = 675.$$

Q - 44

The average weight of P, Q, R and S is 52 kg. A new person T is joined in the group, then the average weight of the group is increased by 2 kg. Again a new person U replaces P, then the average of 5 persons is 55 kg. What is the average weight of Q, R, S and U?

Sol:

$$\text{Sum of 4 persons P, Q, R, S} = 52 \times 4 = 208 \text{ kg} \rightarrow (1)$$

Sum of 5 persons P, Q, R, S, T = $54 \times 5 = 270$ kg $\rightarrow$ (2) $\therefore$ Average is increased by

Sum of 5 persons Q, R, S, T, U = $55 \times 5 = 275$ kg $\rightarrow$ (3) $\therefore$ U replaces with P

(2) – (1) $\Rightarrow$ T = $270 - 208 = 62$ kg substitute in (3)

Q + R + S + 62 + U = 275 $\Rightarrow$ Q + R + S + U = $275 - 62 = 213$ kg

$\therefore$ Average of Q, R, S, U = $\frac{\text{Sum of } Q, R, S, U}{\text{No. of persons}} = \frac{213}{4} = 53.25$ kg.

Q - 45

In the first 20 overs of a cricket match, the run rate is only 4.8. What should be the run rate in the next 30 overs to reach the target of 345 runs?

Sol:

$$\text{Runs scored} = \text{No. of overs} \times \text{Run rate}$$

Runs scored in first 20 overs = $20 \times 4.8 = 96$

Required runs in next 30 overs = Target – Runs in first 20 overs = $345 - 96 = 249$ runs

$\therefore$ Run rate for next 30 overs = $\frac{\text{Required runs}}{\text{No. of overs}} = \frac{249}{30} = 8.3$.

Q - 46

Bumrah's bowling average is 17.5. He took 5 wickets for 25 runs and thereby his average decreases by 0.5. How many wickets taken by him initially?

Sol:

$$\text{Bowling Average} = \frac{\text{Total runs given}}{\text{No. of wickets}}$$

Consider, initial no. of wickets = x

Total runs given = $17.5 \times x = 17.5x$

New bowling average = $\frac{17.5x + 25}{x + 5} = 17.5 - 0.5 = 17$ $\therefore$ Average is decreased by 0.5

$17.5x + 25 = 17x + 25 \Rightarrow 0.5x = 60 \Rightarrow x = 120$

$\therefore$ Initial number of wickets = 120.

Q - 47

The average wickets per test match of a bowler was 6. If 4 matches, where he took no wickets are excluded and he takes 13 wickets per match in a 5 match series, his average increased to 7 wickets per match. How many matches did he play before the latest series?

Sol:

$$\text{Average wickets per match} = \frac{\text{Total no. of wickets}}{\text{No. of matches}}$$

Consider, initial no. of matches = x

Initial total no. of wickets = $6 \times x = 6x$

After exclusion of 4 matches and inclusion of 5 match series,

New average = $\frac{6x - (0 \times 4) + (13 \times 5)}{x - 4 + 5} = 7$ $\therefore$ In 5 match series he took 13 wickets/match

$6x + 65 = 7x + 7 \Rightarrow x = 58$

∴ Initial no. of matches $x = 58$.

Q - 48

The average marks of 42 students are 51 but after checking there are two mistakes found. After adjustment, if a student got 28 marks more and other student got 70 marks less, then what will be the adjusted average?

Sol: Sum of marks of 42 students initially $= 51 \times 42 = 2142$ ∴ Sum = Avg. $\times$ No. of obs.

$$\text{After adjustment average} = \frac{2142 - 28 + 70}{42} = 52$$

(or) ∴ By mistake 1st student got 28 more, 2nd got 70 less marks

$$\therefore \text{Adjusted average} = 51 + \frac{(-28 + 70)}{42} = 52.$$

Q - 49

The average marks of the students in 4 sections A, B, C, D is 60. The average of students of A, B, C, D individually are 45, 50, 72 and 80 respectively. If the average marks of students of section A and B together is 48 and that of B and C together is 60, then what is the ratio of number of students in sections A and D?

Sol: Given that, average of students in A, B, C, D = 60

Also, average of students in B and C = 60

“To maintain overall average is 60, then A and D average also must be 60”.

Consider, no. of students in A = x , no. of students in D = y

Average of section A = 45, Average of section D = 80

$$\text{Average of A and D} = \frac{\text{Sum of A} + \text{Sum of D}}{\text{No. of students in A and D}}$$

$$60 = \frac{45x + 80y}{x + y} \Rightarrow 60x + 60y = 45x + 80y \quad \therefore \text{Sum} = \text{Avg.} \times \text{No. of obs.}$$

$$15x = 20y \Rightarrow x : y = 20 : 15 = 4 : 3$$

∴ Ratio of students in A and D = 4 : 3.

Q - 50

The average score of Virat after 58 innings is 58 and in the 59th innings he scores 117 runs. In the 60th innings, find the minimum number of runs required to increase his average score by 1 than it was before 60th innings.

Sol: Sum of runs after 58 innings $= 58 \times 58 = 3364$

$$\text{Average after 59th innings} = \frac{\text{Sum of 58 innings} + \text{59th innings}}{\text{No. of innings}} = \frac{3364 + 117}{59} = 59$$

$$\text{Average after 60th innings} = \frac{\text{Sum of 58 innings} + \text{59th innings} + \text{60th innings}}{\text{No. of innings}}$$

$$59 + 1 = \frac{3364 + 117 + x}{60} \Rightarrow x = 119 \quad \therefore \text{Average is increased by 1 (59 + 1)}$$

∴ Required minimum no. of runs in 60th innings $x = 119$.

ASSESSMENT TEST

1. Find the average of 120 natural numbers.
2. Find the average of squares of natural numbers from 1 to 53.
3. What is the average of cubes of natural numbers from 1 to 40?
4. If $38a + 38b = 4902$, what is the average of a and b ?
5. a) What is the average of odd numbers from 1 to 95?
b) What is the average of odd numbers from 1 to 126?
6. a) What is the average of even numbers from 1 to 108?
b) What is the average of even numbers from 1 to 135?
7. Find the average of first 75 consecutive odd numbers.
8. Find the average of first 64 consecutive even numbers.
9. Find the average of first 21 multiples of 9.
10. Find the average of multiples of 13 in the natural number series from 200 to 500.
11. The average of 20 numbers is 17. What will be the new average, if each number is multiplied by 7?
12. The average of 9 consecutive even numbers is 44. What is the smallest number?
13. The average of 14 consecutive odd numbers is 64, then find the difference between second smallest and largest numbers.
14. A number 'p' is equal to 60% of the average of 5, 7, 9, 11 and a number 'q'. If the average of p and q is 40, then find the value of q.
15. The average of 8 consecutive even numbers A, B, C, D, E, F, G and H is 37. What is the product of C and G?
16. The average of 'x' numbers is y^2 and the average of 'y' numbers is x^2 . Find the average of all the numbers.
17. The average cost of 'x' articles is Rs. 40 and that of 'y' articles is Rs. 62. The average cost of all articles combinedly is Rs. 54. What is $x : y$?
18. There are 80 boys and 70 girls in a class. If the average contribution of amount for boys is Rs. 68 and that of girls is Rs. 83. What is the average contribution of entire class?
19. Gopal rides a bike from his house at a speed of 60 kmph while going to office and return home at a speed of 40 kmph, then find the average speed during the entire journey.
20. A person covers three equal distances at a speeds of 18 kmph, 24 kmph and 36 kmph. What is the average speed during the entire journey?

21. A man starts from his house and covers 30 km at a speed of 6 kmph, next 24 km at a speed of 8 kmph and last 36 km at a speed of 9 kmph, then find the average speed of a man for the entire journey.
22. A person covers first $\frac{1}{3}$ rd of distance at a speed of 10 kmph, next $\frac{1}{6}$ th of distance at a speed of 15 kmph and the remaining distance at a speed of 20 kmph. Find the average speed during the whole journey.
23. Out of three numbers, second is twice the first and is also four times of third. If the average of three numbers is 91, then find the largest number.
24. The average weight of L and M is 54 kg, that of M and N is 58 kg and the average weight of N and O is 63 kg. Find the weight of N.
25. The average temperature from Sunday to Thursday is 44°C and from Monday to Friday is 47°C. If the temperature on Friday is 53°C, then what is the temperature on Sunday?
26. The average weight of girls in a class is 45 kg and that of boys is 54 kg. If the ratio of number of boys and girls is 5 : 4, then find the average weight of students in the class.
27. The average of 'n' observations is 73. If one more observation of value 139 is added, then the average is increased by 1. Find the value of 'n'.
28. A company produces on an average 5900 articles per month for the first 5 months. How many articles it must produce on an average per month over the next 7 months, to maintain the overall average of 5375 articles per month?
29. Total strength of the class is 180. The average marks of boys is 39 and that of girls is 75. If the average marks of entire class is 60, then find the number of girls in the class.
30. The average salary of entire staff in a college is Rs. 650 per day. The average salary of teaching staff is Rs. 800 and that of non-teaching staff is Rs. 230. Find the number of teaching staff, if the number of non-teaching staff is 60?
31. Twelve friends went to shopping for buying clothes. Eleven of them spent Rs. 2500 on their clothes and the twelfth one spent Rs. 550 more than the average expenditure of all twelve. What is the money spent by twelfth person?
32. The average weight of 41 children is 30 kg. The average of first 20 of them is 31 kg and that of last 20 is 28 kg, then find the weight of 21st child.
33. The average of 23 results is 70. If the average of first 12 results is 67 and that of last 12 results is 71. What is twelfth result?
34. The average of 30 observations is 26. Later it was found that an observation 20 was wrongly entered as 80, then find the new average.

35. The average expenditure on 12 articles is decreased by 0.5 when the cost of one of the article Rs. 50 is replaced by another. Find the cost of new article.
36. The average price of 10 pairs of clothes is Rs. 1950. If the price of trolley bag is also included, then the average is increased by Rs. 50. What is the price of trolley bag?
37. A batsman makes a score of 94 runs in the 15th innings and thus his average is increases by 3 runs. What is his average after 15th innings?
38. The average score of a batsman in 16 innings is 43. If the highest and lowest scores were excluded, then the average of remaining innings would decrease by 3. If the difference between these two innings is 62, then find the lowest score.
39. There are 63 students who are going to mess. If 9 new students are admitted, the average expenditure of the mess is increased by Rs. 108 per day and the average expenditure per head is decreased by 1 rupee. What is the original expenditure of the mess?
40. The average weight of 35 students in a class is 48 kg. 5 of them whose average weight is 45 kg leave the class and other 5 students whose average weight is 50 kg join the class. What is the new average weight of the class?
41. In an exam, a student average marks were 74 per paper. If he had obtained 25 marks more in Maths paper and 17 marks more for his Science paper, his average per paper would have been 77. How many papers were there in the exam?
42. The average height of 3 men P, Q and R is 172 cm. Another man S joins the group and the average now becomes 170 cm. If another man T, whose height is 6 cm more than that of S, replaces P. Now the average height of Q, R, S and T becomes 169 cm. Find the height of P.
43. The average marks of 35 students is 47. But after checking, there are 2 mistakes found. After adjustment, if a student got 60 marks more and other student got 25 marks less. What will be the adjusted average?
44. Murali's bowling average is 18.2. He took 6 wickets for 30 runs and there by his average decreases by 0.2. How many wickets taken by him initially?
45. The average wickets per test match of a bowler was 4. If two matches, where he took no wickets are excluded and he takes 10 wickets per match in a 4 match series, his average increases to 5 wickets per match. How many matches did he play before the latest series?

KEY

1. 60.5
2. 963
3. 16810
4. 64.5
5. a) 48 b) 63
6. a) 55 b) 68
7. 75
8. 65
9. 99
10. 351
11. 119
12. 36
13. 24
14. 68
15. 1428
16. xy
17. 4 : 7
18. Rs. 75
19. 48 kmph
20. 24 kmph
21. 7.5 kmph
22. 14.4 kmph
23. 52
24. 67 kg
25. 38°C
26. 50 kg
27. 65
28. 5000
29. 105
30. 168
31. Rs. 3100
32. 50 kg
33. 46

- 34. 24
- 35. Rs. 44
- 36. Rs. 2500
- 37. 52
- 38. 33
- 39. Rs. 1260
- 40. $48\frac{5}{7}$ kg
- 41. 14
- 42. 174 cm
- 43. 46
- 44. 390
- 45. 30